Lorentz – Drude model

$$m\left(\ddot{\vec{r}} + \underset{\substack{\uparrow \\ \text{damping} \\ \text{force}}}{\gamma \dot{\vec{r}}} + \omega_0^2 \vec{r}\right) = q\vec{E}(r,t) \qquad (q=-e)$$

Assume wave length $\lambda \gg$ typical displacement $\vec{r}$

$$\Rightarrow \vec{E}(r,t) = \vec{E}(\omega)\, e^{-i\omega t} \quad \Rightarrow \vec{r}(t) = \vec{r}(\omega)\, e^{-i\omega t}$$

$$\left(-\omega^2 - i\omega\gamma + \omega_0^2\right)\vec{r}(\omega) = \frac{q}{m}\vec{E}(\omega)$$

$$\vec{p}(\omega) = q\vec{r}(\omega) = \frac{q^2/m}{\omega_0^2 - \omega^2 - i\omega\gamma}\vec{E}(\omega) \Rightarrow \vec{P} = N\vec{p}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0\left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}\right)$$

where $N = \#$ number of molecules

$Z = \#$ electrons per molecule

<u>Insulators</u>: the lowest resonant frequency is finite, $\omega_j \neq 0$

$$\Rightarrow \frac{\varepsilon(\omega)}{\varepsilon_0} \to \left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_j \frac{f_j}{\omega_j^2}\right)$$

<u>Conductors</u> some fraction, f_0, of electrons are free

$\Rightarrow$ for them $\omega_j = 0$ (no restoring force)

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0\left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_{j\neq 0}\frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}\right) +$$

$$\varepsilon_0 \frac{Ne^2 f_0}{\varepsilon_0 m\omega(\omega + i\gamma)} = \varepsilon_b(\omega) + \frac{i\sigma(\omega)}{\omega}$$

$$\Rightarrow i\sigma(\omega) = -\frac{Ne^2 f_0}{m(\omega+i\gamma)} = -\frac{Ne^2 f_0}{i\gamma m\left(1-\frac{i\omega}{\gamma}\right)} = i\,\frac{Ne^2 f_0 \tau}{m}\cdot\frac{1}{1-i\omega\tau}$$

$$\gamma = \frac{1}{\tau}$$

Thus obtain $\sigma(\omega) = \dfrac{\sigma_0}{1-i\omega\tau}$ (Drude's result)

where $Nf_0 \equiv n$ = density of free electrons.

Two important limits:

(a) low-frequency (d.c.)

$$\varepsilon(\omega) \to \varepsilon_b(0) + \frac{i\sigma_0}{\omega} \approx \frac{i\sigma_0}{\omega}$$ dominated by $\frac{\sigma_0}{\omega}$ term.

(b) high-frequency

$$\varepsilon(\omega) \to \varepsilon_b(\omega) - \varepsilon_0\frac{Nf_0 e^2}{\varepsilon_0 m\omega^2} = \varepsilon_b(\omega) - \varepsilon_0\frac{\omega_0^2}{\omega^2}$$

where $\omega_0^2 = \dfrac{Nf_0 e^2}{\varepsilon_0 m} = \dfrac{ne^2}{\varepsilon_0 m}$ = plasma frequency.

$\varepsilon(\omega)$ and absorption

$$\vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \cancel{\chi(\omega)}\, \vec{P}(\omega) = \varepsilon_0 \left(1+\chi(\omega)\right) \vec{E}(\omega) = \varepsilon(\omega)\, \vec{E}(\omega)$$

$$\hookrightarrow \varepsilon_0 \chi(\omega) \vec{E}(\omega)$$

Dielectric function $\varepsilon(\omega) = \varepsilon_0 \left(1+\chi(\omega)\right)$

Hence $\varepsilon(\omega)$ has the same properties as $\chi(\omega)$:

$$\varepsilon'(\omega) = \varepsilon'(-\omega) \quad , \quad \varepsilon''(\omega) = -\varepsilon''(-\omega).$$

From $\vec{P} = \vec{D} - \varepsilon_0 \vec{E} = (\varepsilon - \varepsilon_0)\, \vec{E}$, $\vec{E}(t) = \vec{E}(\omega)\, e^{-i\omega t}$

$$\frac{\partial \vec{P}}{\partial t} = \vec{J} = \left(\frac{\partial}{\partial t} \frac{\sum q \vec{r}}{\Delta V} = \frac{\sum q \vec{v}}{\Delta V} \right)$$

$$\Rightarrow -i\omega \vec{P}(\omega) = -i\omega \left(\varepsilon(\omega) - \varepsilon_0\right) \vec{E}(\omega)$$

$$\Rightarrow -i\omega \vec{P}(\omega) = \underbrace{\vec{J}(\omega) = \sigma(\omega)\, \vec{E}(\omega)}_{\text{Ohm's law for conductors}}$$

$$\Rightarrow \vec{P}(\omega) = \left(\varepsilon(\omega) - \varepsilon_0\right) \vec{E}(\omega) = \frac{i\sigma(\omega)}{\omega} \vec{E}(\omega)$$

$$\varepsilon(\omega) = \varepsilon_0 + \frac{i\sigma(\omega)}{\omega}.$$

Connection with energy absorption — Joule heat

power $P = \vec{J} \cdot \vec{E}$

$$\vec{E}(t) = \mathrm{Re}\left\{ \vec{E}(\omega)\, e^{-i\omega t} \right\} = \frac{1}{2} \left(\vec{E}(\omega)\, e^{-i\omega t} + \vec{E}^* e^{i\omega t} \right)$$

$$\vec{J}(t) = \mathrm{Re}\left\{ \underset{\sigma(\omega)\vec{E}(\omega)}{\underset{\parallel}{\vec{J}(\omega)}}\, e^{-i\omega t} \right\} = \frac{1}{2} \left(\sigma(\omega)\, \vec{E}(\omega)\, e^{-i\omega t} + \sigma^*(\omega)\, \vec{E}^*(\omega)\, e^{i\omega t} \right)$$

$$P = \frac{1}{4}\left(\vec{E}_\omega e^{-i\omega t} + \vec{E}_\omega^* e^{i\omega t}\right)\left(\sigma(\omega)\vec{E}_\omega e^{-i\omega t} + \sigma^*(\omega)\vec{E}_\omega^* e^{i\omega t}\right)$$

$$= \frac{1}{4}\Big(\sigma(\omega)\, \vec{E}_\omega \cdot \vec{E}_\omega e^{-i2\omega t} + \sigma^*(\omega)\vec{E}_\omega^* \cdot \vec{E}_\omega^* e^{i2\omega t}$$

$$+ \left(\sigma(\omega) + \sigma^*(\omega)\right) \vec{E}_\omega \cdot \vec{E}_\omega^* \Big)$$

Average power over the period $\Rightarrow \overline{e^{\pm i2\omega t}} = 0$.

$$\overline{P} = \frac{1}{2}\sigma'(\omega)\, |\vec{E}(\omega)|^2 = \frac{1}{2}\omega\, \varepsilon''(\omega)\, |\vec{E}(\omega)|^2.$$

$$\varepsilon^*(\omega) = \varepsilon_0 + \frac{i}{\omega}\left(\sigma'(\omega) + i\sigma''(\omega)\right)$$

$$\Rightarrow \begin{cases} \varepsilon'(\omega) = \varepsilon_0 - \dfrac{\sigma''(\omega)}{\omega} \\ \varepsilon''(\omega) = \dfrac{\sigma'(\omega)}{\omega} \end{cases}$$

$$\overline{P}(\omega) > 0 \Rightarrow \begin{cases} \varepsilon''(\omega) > 0 & \text{for } \omega > 0 \\ \varepsilon''(\omega) < 0 & \text{for } \omega < 0. \end{cases}$$

Drude model: $\sigma(\omega) = \dfrac{ie^2 n}{m\left(\omega + \frac{i}{\tau}\right)} = \dfrac{e^2 n \tau}{m(1 - i\omega\tau)} = \dfrac{\sigma_0}{1 - i\omega\tau}$

$$\Rightarrow \sigma^*(\omega) = \frac{\sigma_0(1 + i\omega\tau)}{1 + \omega^2\tau^2} \longrightarrow \begin{cases} \sigma_0, & \omega\tau \ll 1 \text{ large absorption} \\ \dfrac{i\sigma_0}{\omega\tau}, & \omega\tau \gg 1 \text{ low absorption} \end{cases}$$